

Research Progress Report:
Effective Rank as a Diagnostic for Representation Bottlenecks and
Compression in Language Models

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Part I

Research Plan from the Proposal

1 Motivation and Background

Large language models (LLMs) follow a common design pattern: they are built in an overparameterized fashion and subsequently compressed after training, most often through post-training quantization or low-rank factorization. This practice raises a fundamental question: *how much of the parameter space is actually utilised by a trained model, in both its intermediate representations and its weight matrices?*

If a trained network uses far fewer effective directions than its nominal dimensionality would suggest, then the spectra of its representations and weights should reveal this. Our project proposes **effective rank**—the exponential of the spectral entropy of the singular-value distribution, as defined by Roy & Vetterli (2007) [1]—as a lightweight, layer-level diagnostic that can simultaneously:

- (a) reveal *representation bottlenecks* (layers whose internal dimensionality is compressed),
- (b) predict *compressional potential* (which layers can tolerate aggressive low-rank or low-bit approximation), and
- (c) connect to the *Information Bottleneck* (IB) framework by serving as a tractable proxy for the mutual information complexity term $I(T; X)$ that is otherwise intractable in high-dimensional continuous spaces.

1.1 Connection to Information Theory and Cognition

This project is situated within the course “Information Theory and Cognition,” whose central framework is the Information Bottleneck (IB) method developed by Tishby, Pereira & Bialek (1999) [2]. The IB formalises optimal compression: given an input X and a relevant variable Y , find a compressed representation T that minimises the complexity $I(T; X)$ while maximising the informativeness $I(T; Y)$:

$$\mathcal{L}_{\text{IB}} = I(T; X) - \beta \cdot I(T; Y).$$

Tishby & Zaslavsky (2015) [3] argued that each layer of a deep neural network can be viewed as an IB representation, with optimal architectures corresponding to bifurcation points on the IB tradeoff curve.

The key bridge is provided by the “Inductive Bottleneck” paper (arXiv:2512.07331) [4], which uses the Effective Encoding Dimension (EED)—defined identically to Roy & Vetterli’s effective rank—as a tractable proxy for $I(T; X)$ in Vision Transformers. Their central finding is a **U-shaped effective rank profile** across layers: representations compress in middle layers and expand in later layers, mirroring the compression phase predicted by IB theory.

Our project asks: *do language models, trained to model human language, develop analogous bottleneck-structured representations?* This connects to several cognitive science concepts:

- **Efficient coding hypothesis** (Barlow, 1961) [5]: neural systems encode sensory information by reducing statistical redundancy. Low effective rank in learned representations implies efficient codes.

- **Cognitive bottlenecks:** biological cognition operates under severe resource constraints—limited attention, bounded working memory (~ 4 items). These bottlenecks force the brain to extract maximally relevant compressed representations. Post-training compression of LLMs is an artificial analogue: which layers can tolerate aggressive “cognitive” compression?
- **Language as compression** (Zaslavsky et al., 2018) [6]: human languages achieve near-optimal compression on the IB tradeoff curve when mapping continuous meaning spaces into finite lexicons. An LLM trained on language should reflect this compression structure internally.

2 Concrete Research Plan

The proposal specifies the following experimental pipeline, centred on a single layer of a neural network:

$$y = \phi(Wu), \quad u = \text{Norm}(x),$$

where x is the activation from the previous layer, u is the normalised activation, W is the weight matrix, and ϕ is the nonlinearity.

2.1 Step 1: Model Selection

Choose small language models that are computationally tractable. The original proposal targeted Qwen 3 0.6B; the executed experiments expanded to four models.

2.2 Step 2: Data Collection

Collect normalised activations $[u_1, u_2, \dots]$ from calibration data (WikiText-2 training set) and stack them into a matrix U . Also collect the trained weight matrices W for each layer.

2.3 Step 3: Spectral Decomposition

Compute the spectra of both the activation covariance and the weight matrices:

$$U^T U = Q \Lambda Q^T, \quad W = U \Sigma V^T,$$

yielding eigenvalues $\{\lambda_i\}$ and singular values $\{\sigma_i\}$.

2.4 Step 4: Effective Rank Computation

Compute spectral entropy and effective rank:

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j} \quad \text{or} \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}, \quad H(\mathbf{p}) = - \sum_i p_i \log p_i, \quad r_u \text{ or } r_w = e^{H(\mathbf{p})},$$

where r_u is the effective rank of the representation spectrum (intrinsic dimensionality of the input) and r_w is the effective rank of the weight spectrum (intrinsic capacity of the operator).

2.5 Step 5: Diagnosis and Compression

Use r_u and r_w to guide diagnosis and compression:

- A large r_u with a small r_w may indicate a *bottleneck*.
- A small r_u with a small r_w may indicate a well-matched, compressible layer.
- A small r_u with a large r_w suggests the weights can be compressed.
- Test direct low-rank factorization: $Wu \approx WV_{r_u}V_{r_u}^T u$.
- Test allocating quantisation bits based on effective rank (low $r_{\text{eff}} \rightarrow$ lower bits).

2.6 Step 6: Evaluation

Compare compressed models against the uncompressed baseline using perplexity on the WikiText-2 test/validation set, and visualise how r_u and r_w change across layers to identify structural patterns.

Part II

Experimental Implementation and Findings

This part describes the complete experimental pipeline implemented by Peng Zhao, including the logic behind each stage, the design choices made, and the empirical results obtained.

3 Pipeline Overview

The experiments are organised as a six-stage pipeline (stages 0–5), implemented as standalone Python scripts that communicate through well-defined file interfaces. The pipeline was executed across **four models**:

1. **Qwen3-0.6B** — 28 layers, hidden size 1024, intermediate size 3072
2. **Llama-3.2-1B** — 16 layers, hidden size 2048, intermediate size 8192
3. **Qwen2.5-1.5B** — 28 layers, hidden size 1536, intermediate size 8960
4. **SmolLM2-1.7B** — 24 layers, hidden size 2048, intermediate size 8192

All models were evaluated on WikiText-2 with identical tokenisation and batching procedures.

4 Stage 0: Data and Model Setup

Two simple download scripts (`0_download_model.py`, `0_download_data.py`) handle:

- Downloading each Hugging Face base model into `artifact/models/<model_id>/base`.
- Downloading the WikiText-2 corpus into `data/wikitext2/` (train, validation, test splits).

The design choice here is intentional minimalism: keep setup completely separate from analysis.

5 Stage 1: Collecting Layer-wise Statistics

`1_collect_statistics.py` is the core data-collection script. It performs a single forward pass over the calibration data (up to 65,536 token activations per layer from WikiText-2 training set, with sequence length 512) and collects the following quantities for each of the 28 (or 16/24) transformer layers:

5.1 Compression Scope

A critical design decision: the project compresses only the MLP `gate_proj` and `up_proj` weight matrices. The MLP `down_proj` and all attention projections are left uncompressed. This focuses the analysis on the two parallel linear projections that feed the SwiGLU activation.

5.2 Activation Locations

Three activation locations are monitored per layer:

- **MLP input:** output of `post_attention_layernorm` (the normalised activation seen by `gate_proj` and `up_proj`).
- **MLP output:** output of `layer.mlp` (the contribution added to the residual stream).
- **Residual stream:** input to `post_attention_layernorm` (the pre-normalisation residual, used only as a diagnostic reference).

5.3 Statistics Collected

For each layer, the script computes:

- **Weight effective rank** for `gate_proj` and `up_proj`: computed from singular values using $p_i = \sigma_i / \sum_j \sigma_j$, then $r_w = \exp(H(\mathbf{p}))$. The mean of the two weight effective rank ratios ($r_w / \min(m, n)$) serves as the layer-level weight rank summary.
- **Activation effective rank (centered):** computed from eigenvalues of the empirical covariance matrix $\text{Cov}(u) = \mathbb{E}[uu^T] - \mathbb{E}[u]\mathbb{E}[u]^T$. Both MLP input and MLP output centered effective ranks are computed.
- **Activation effective rank (uncentered):** computed from eigenvalues of the uncentered second moment $\mathbb{E}[uu^T]$. This variant captures the mean component as well.
- **MLP output uncentered trace:** $\text{tr}(\mathbb{E}[xx^T])$, equivalently the per-token mean squared norm $\mathbb{E}[\|x\|^2]$. The natural logarithm of this quantity serves as the **main importance/energy signal**.
- **Residual stream uncentered trace:** retained only as a diagnostic reference because the residual stream magnitude is path-dependent and can be distorted by earlier outlier layers.

5.4 Important Interpretation Choices

- **Centered** activation effective rank is the main rank-based signal (dimensionality of the *spread* of activations, ignoring the mean).

- **Log MLP output uncentered trace** is the main importance signal (how strongly a layer writes to the residual stream).
- Residual-relative trace is *not* used as the primary cross-layer importance metric.

5.5 Cached Artefacts

The script saves second-moment matrices (`mlp_input_second_moments.pt`, `mlp_output_second_moments.pt`), eigenvalue spectra (`activation_eigenvalues.pt`), weight singular values (`weight_singular_values.pt`), and a summary CSV (`layer_statistics.csv`). These caches are reused by downstream stages without re-running inference.

6 Stage 2: Descriptive Visualisation

`2_plot_statistics.py` produces three families of plots per model:

1. **Compressibility profile:** layer-wise plots of MLP input centered e-rank ratio, MLP output centered e-rank ratio, MLP input/output uncentered e-rank ratio, and mean weight e-rank ratio.
2. **Importance profile:** layer-wise plot of MLP output log trace (main signal) and residual-stream log trace (diagnostic reference).
3. **Compressibility–importance overlay:** the mean of input-centered and output-centered e-rank ratios (“compressibility summary”) overlaid with min-max normalised MLP output log trace (“importance summary”).

Stage 2 is purely descriptive; it does not compute any budget or decision score.

7 Stage 3: Rank Budget Computation and Compression

This is the central experimental contribution, implemented across two scripts.

7.1 Rank Budget Computation (`3_compute_rank_budgets.py`)

The script converts the stage-1 statistics into a layerwise rank-budget proposal via a simple, interpretable formula.

Rank signal.

$$\text{rank_signal}_\ell = \frac{1}{2} \left(\frac{r_{u,\text{in}}^{(\ell)}}{d} + \frac{r_{u,\text{out}}^{(\ell)}}{d_{\text{out}}} \right),$$

where $r_{u,\text{in}}^{(\ell)}$ and $r_{u,\text{out}}^{(\ell)}$ are the centered activation effective ranks of the MLP input and output, and d, d_{out} are their respective feature sizes. This is then min-max normalised across layers.

Importance signal.

$$\text{importance_signal}_\ell = \text{min-max-normalise} \left(\log \text{tr} \left(\mathbb{E} [x_\ell x_\ell^T] \right) \right).$$

Protection score.

$$\text{protection_score}_\ell = w \cdot \text{rank_signal}_\ell + (1 - w) \cdot \text{importance_signal}_\ell,$$

where $w \in [0, 1]$ is a tunable weight. Higher protection score $\Rightarrow$ more rank budget (the layer is either structurally complex or writes strongly to the residual stream).

Budget allocation.

1. Choose compression ratio α : the target factorised/dense parameter ratio.
2. Derive the uniform rank implied by α : $k_{\text{uniform}} = \alpha \cdot \frac{d_{\text{hidden}} \cdot d_{\text{intermediate}}}{d_{\text{hidden}} + d_{\text{intermediate}}}$.
3. Redistribute the same total budget $k_{\text{uniform}} \times L$ across layers proportionally to $\text{protection_score}_\ell$.
4. Enforce integer constraints and the upper bound $k \leq d_{\text{hidden}}$.
5. Adjust rounding so the final layerwise total matches the uniform total exactly.

Experimental grid. The experiments swept over:

- $\alpha \in \{0.50, 0.80\}$ (50% and 80% of dense parameters retained)
- $w \in \{0.25, 0.50, 0.75\}$

7.2 Model Compression (3_compress_models.py)

For each budget setting, three compression methods are applied:

Plain SVD Uniform. For each target layer, compute the truncated SVD $W \approx U_k \Sigma_k V_k^T$ and replace the weight with the rank- k approximation. The same uniform rank k_{uniform} is used for every layer.

Activation-Aware Uniform. Following the ASVD (Activation-aware SVD) paradigm:

1. Compute the empirical second moment of MLP inputs: $S = \mathbb{E}[uu^T]$.
2. Compute $S^{1/2}$ and $S^{-1/2}$ (with eigenvalue floor 10^{-8}).
3. Form the transformed weight: $\widetilde{W} = W \cdot S^{1/2}$.
4. Compute truncated SVD of $\widetilde{W}$ at rank k .
5. Reconstruct: $W_k^{\text{ASVD}} = \widetilde{U}_k \widetilde{\Sigma}_k \widetilde{V}_k^T \cdot S^{-1/2}$.

This ensures that the low-rank approximation minimises the *activation-weighted* reconstruction error $\|Wx - W_k x\|$ rather than the raw Frobenius error $\|W - W_k\|_F$. The same uniform rank is used for all layers.

Activation-Aware Layerwise. Same activation-aware SVD procedure, but now each layer receives its own rank from the stage-3 budget allocation. This is the method that tests the core hypothesis: *can effective-rank-guided layerwise allocation outperform uniform allocation?*

8 Stage 4: Perplexity Benchmarking

`4_benchmark_perplexity.py` evaluates every compressed variant and the base model on the WikiText-2 validation set using fixed token windows (length 512, up to 256 windows, batch size 4). It computes:

$$\text{Perplexity} = \exp \left(\frac{1}{N} \sum_{i=1}^N -\log p(x_i \mid x_{<i}) \right),$$

where N is the total number of predicted tokens.

9 Stage 5: Cross-Model Analysis

Two analysis scripts produce cross-model comparisons:

9.1 Layer Analysis (`5_layer_analysis.py`)

Produces two key plots:

1. **Rank–trace scatter plot:** normalised rank signal vs. normalised trace signal across all models and layers, with per-model and pooled Pearson correlation.
2. **Rank–trace trend plot:** normalised rank and trace signals plotted against normalised layer depth, revealing structural trends (early/middle/late regions).

9.2 Compression Analysis (`5_compression_analysis.py`)

Produces:

1. Per-model comparison plots: activation-aware uniform (horizontal baseline) vs. activation-aware layerwise (varying with w), with Δ -NLL annotations.
2. Cross-model aggregate plot: layerwise–uniform delta NLL vs. w .
3. Cross-model heatmap: delta NLL by model and w .

10 Key Experimental Results

10.1 Finding 1: Strong Negative Rank–Trace Correlation

Across all four models, there is a consistent **strong negative correlation** between the normalised rank signal (activation effective rank ratio) and the normalised trace signal (MLP output energy):

Table 1: Pearson correlation between normalised rank signal and normalised trace signal.

Model	Layers	Pearson r
Llama-3.2-1B	16	−0.72
Qwen2.5-1.5B	28	−0.73
Qwen3-0.6B	28	−0.76
SmolLM2-1.7B	24	−0.74
All pooled	96	−0.69

Interpretation. Layers that have low effective rank (concentrated spectrum, fewer effective dimensions) tend to write more strongly to the residual stream (higher trace). This suggests a *rank–importance tradeoff*: the most “important” layers (highest energy contribution) are precisely those that operate in the most compressed subspace. This is consistent with the IB perspective: layers that have compressed their representations the most ($\downarrow I(T; X)$) while maintaining high downstream contribution ($\uparrow I(T; Y)$) sit at efficient points on the IB tradeoff curve.

10.2 Finding 2: Activation-Aware SVD Massively Outperforms Plain SVD

Plain truncated SVD is catastrophically poor for LLM compression. Activation-aware SVD recovers most of the quality:

Table 2: Perplexity comparison at $\alpha = 0.80$ (80% parameter retention for gate/up projections).

Method	Qwen3-0.6B	Llama-3.2-1B	Qwen2.5-1.5B	SmolLM2-1.7B
Base (dense)	31.9	13.4	13.1	10.6
Plain SVD uniform	2980	725	734	289
Activation-aware uniform	45.8	27.7	18.9	18.0

The plain SVD is 50–100 $\times$ worse than the base model, while activation-aware SVD preserves reasonable quality (1.4–2.1 $\times$ base perplexity at $\alpha = 0.80$). This validates the roadmap’s recommendation to use ASVD-style compression rather than naive SVD truncation.

10.3 Finding 3: Layerwise Allocation—Honest, Mixed Results

The central question—can effective-rank-guided layerwise allocation beat uniform allocation?—yields a nuanced answer:

Table 3: Activation-aware perplexity: uniform vs. best layerwise setting at $\alpha = 0.80$.

Model	Uniform	Best Layerwise	Best w
Qwen3-0.6B	45.8	52.8	0.50
Llama-3.2-1B	27.7	163.7	0.25
Qwen2.5-1.5B	18.9	20.0	0.50
SmolLM2-1.7B	18.0	132.4	0.75

At $\alpha = 0.80$, the layerwise allocation consistently fails to beat the uniform baseline. For Llama-3.2-1B and SmolLM2-1.7B, the layerwise variants are dramatically worse.

At $\alpha = 0.50$ (more aggressive compression), the picture is more mixed:

Table 4: Activation-aware perplexity at $\alpha = 0.50$: uniform vs. layerwise.

Model	Uniform	$w=0.25$	$w=0.50$	$w=0.75$
Qwen3-0.6B	103.4	188.7	121.2	111.3
Llama-3.2-1B	258.5	3361.8	2952.5	2611.0
Qwen2.5-1.5B	64.7	947.7	154.7	110.0
SmolLM2-1.7B	143.9	4034.3	2049.8	1371.4

At $\alpha = 0.50$, the layerwise allocation is strictly worse than uniform for all models and all w settings.

10.4 Finding 4: Layer-2 Anomaly in Qwen3-0.6B

The Qwen3-0.6B layer statistics reveal a striking anomaly at layer 2: the MLP output effective rank drops to ~ 1.0 (ratio ≈ 0.001) while the MLP output uncentered trace spikes to $\sim 90,318$ ($\log \approx 11.4$). This means layer 2’s MLP output is essentially rank-1 but with enormous magnitude—a classic “outlier layer” that concentrates all its energy in a single direction. Subsequent layers (3+) show the trace dropping back to ~ 47 – 80 , suggesting the residual stream has absorbed this spike.

10.5 Summary of the Pipeline’s Logic

The thinking logic behind the entire pipeline can be summarised as:

1. **Measure two orthogonal signals:** rank structure (how complex is the layer’s representation?) and energy/importance (how strongly does the layer write to the residual stream?).
2. **Combine them:** a layer that is both structurally complex (high rank) and energetically important (high trace) deserves more rank budget (higher “protection”).
3. **Test the hypothesis honestly:** compare the layerwise allocation against a uniform baseline using the same total rank budget, so any improvement must come from *allocation* rather than *total capacity*.
4. **Accept negative results:** the pipeline is designed to “evaluate honestly,” not to force a win. The pipeline’s own design notes state: “v3 accepts that complicated layerwise allocation may not always beat a uniform budget.”

Part III

Limitations, Evaluation, and Future Directions

11 Honest Evaluation of Current Limitations

11.1 Limitation 1: Layerwise Allocation Consistently Underperforms Uniform

The most significant finding is a **negative result**: the effective-rank-guided layerwise budget allocation does not outperform the simple uniform baseline on any model at any compression ratio tested. This is not a minor margin—for some models (Llama-3.2-1B, SmolLM2-1.7B), the layerwise allocation is *catastrophically* worse.

Possible explanations.

1. **The protection score conflates two conflicting signals.** The rank signal says “this layer is complex, give it more rank.” The importance signal says “this layer writes a lot of energy, protect it.” But the strong negative correlation ($r \approx -0.7$) means these two signals partially cancel: high-importance layers have low rank, so boosting them via importance but

penalising them via rank creates an inconsistent allocation. The w parameter cannot resolve this tension because the signals are *anti-correlated* by nature.

2. **The last layers are outliers in energy but critical for output quality.** Late layers in all models show rapidly increasing trace (output energy) but moderate or declining rank. The layerwise budget may under-allocate rank to these critical layers at low w (importance-dominated) or over-allocate to middle layers at high w (rank-dominated).
3. **Effective rank is a scalar summary of the full spectrum.** Two layers can have the same effective rank but very different spectral shapes (e.g., one with a smooth exponential decay, another with a sharp knee). The layerwise budget treats them identically, potentially misallocating rank.
4. **Error propagation across layers.** The per-layer analysis treats each layer independently, but compression errors propagate through the residual stream. A small error in an early layer can amplify through subsequent layers. Uniform allocation may be more robust to this cascading effect.

11.2 Limitation 2: Only MLP Gate/Up Projections Are Compressed

The current pipeline compresses only 2 of the 7 sub-modules per transformer block (gate_proj and up_proj out of Q, K, V, O, gate, up, down). This means:

- The attention mechanism (which is known to have significant low-rank structure, especially in GQA architectures) is left untouched.
- The MLP down_proj (which maps back from intermediate to hidden size) is not compressed, even though its activation statistics are influenced by the gate/up compression.
- The overall model compression is modest: even at $\alpha = 0.50$, only the gate/up projections are halved, which reduces total parameters by far less than 50%.

11.3 Limitation 3: No Quantisation Experiments

The original proposal planned to test quantisation-bit allocation guided by effective rank (e.g., Q3 for low-rank layers, Q5 for high-rank layers). This was not implemented. The current experiments only test low-rank SVD factorisation. Quantisation is a fundamentally different compression mechanism that may interact differently with effective rank.

11.4 Limitation 4: Single Calibration Dataset

All experiments use WikiText-2 for both calibration (collecting activation statistics) and evaluation (measuring perplexity). This raises the concern that the statistics, and therefore the compression decisions, are tuned to WikiText-2’s distribution. It is unknown whether the effective rank profiles would generalise to code, conversation, or multilingual data.

11.5 Limitation 5: No Per-Layer Sensitivity Analysis

The roadmap called for per-layer quantisation sensitivity analysis (ΔPPL_ℓ : quantise one layer at a time, measure perplexity degradation). This “ground truth” sensitivity was never computed, so we cannot evaluate how well effective rank *predicts* per-layer compression sensitivity. Without this,

we cannot answer the proposal’s core question: is effective rank a good *predictor* of compressional potential?

11.6 Limitation 6: σ vs. σ^2 Asymmetry

As noted in the roadmap’s critical review, the weight effective rank r_w is computed from singular values ($p_i = \sigma_i / \sum_j \sigma_j$), while the activation effective rank r_u is computed from covariance eigenvalues ($p_i = \lambda_i / \sum_j \lambda_j$, where $\lambda_i = \sigma_i^2$). These are fundamentally different normalisations that are not directly comparable, yet the pipeline combines them into a single protection score.

11.7 Limitation 7: No Mathematical Proof Completed

The proposal promised a “mathematical proof of effective rank implying good compression performance.” The roadmap sketched three theorems (low-rank approximation error bound, rate-distortion connection, effective rank as IB complexity proxy), but none were formally completed in the implementation.

12 Future Directions

12.1 Direction 1: Per-Layer Sensitivity Analysis

The single most impactful extension would be to compute the per-layer compression sensitivity ΔPPL_ℓ : compress one layer at a time and measure the perplexity degradation. This provides the “ground truth” against which to evaluate effective rank as a predictor. Key analyses:

- Compute Spearman and Pearson correlation between $r_u^{(\ell)}$, $r_w^{(\ell)}$, $\text{trace}^{(\ell)}$, and ΔPPL_ℓ .
- Compare effective rank against established sensitivity metrics: Hessian trace, activation kurtosis, outlier magnitude.
- Generate scatter plots of each metric vs. ΔPPL_ℓ to visualise predictive power.

12.2 Direction 2: Better Budget Allocation Strategies

The current additive protection score may be too simplistic. Alternatives to explore:

- **Importance-only allocation:** since the rank-importance anti-correlation undermines the additive score, try allocating rank *purely* based on importance (trace), ignoring the rank signal entirely.
- **Multiplicative score:** $\text{protection} = \text{rank_signal}^w \cdot \text{importance_signal}^{1-w}$ (geometric mean), which behaves differently under anti-correlation.
- **Spectral-entropy threshold:** instead of using the scalar effective rank, use the full cumulative spectral entropy to set per-layer truncation points (as in ARSVD [15]).
- **Oracle-guided allocation:** use the per-layer sensitivity ΔPPL_ℓ to derive the optimal allocation, then measure how close effective-rank-based allocation comes to this oracle.

12.3 Direction 3: Information-Theoretic Analysis

To strengthen the IB connection:

- Plot an “information plane” analogue: $(r_u^{(\ell)}, \text{downstream contribution}_\ell)$ for each layer, and check if this reveals an IB-like tradeoff.
- Test whether the U-shaped effective rank profile observed in Vision Transformers (Inductive Bottleneck paper) also appears in language models. The current data hints at a *monotonically* decreasing rank profile in late layers, which may reflect a fundamentally different architecture.
- Formally prove (under Gaussian assumptions) the rate-distortion bound: $R(D) \leq \frac{r_{\text{eff}}^{(\lambda)}}{2} \cdot \log \frac{\bar{\lambda}}{D}$, connecting effective rank to minimum bits needed.

12.4 Direction 4: Expanding Compression Scope

- **Attention projections:** apply the same spectral analysis and SVD compression to Q, K, V, O projections. GQA architectures (Llama-3.2) may show particularly interesting rank structure.
- **MLP down_proj:** complete the MLP compression by also factorising the down projection.
- **Quantisation experiments:** implement the originally proposed effective-rank-guided bit allocation (Q3/Q4/Q5 assignment based on layer effective rank) using GPTQ or AWQ as backends.

12.5 Direction 5: Scale and Generalisation

- Test on larger models (7B+ parameters) where overparameterisation is more pronounced and effective rank patterns may be more distinct.
- Evaluate on diverse calibration datasets (code, conversation, multilingual) to test whether effective rank profiles are distribution-invariant.
- Compare findings across model families to test universality of the rank–trace anti-correlation.

12.6 Direction 6: Cognitive Science Framing

- Frame the observed rank–trace anti-correlation as evidence of efficient coding: layers that contribute the most “information” to downstream processing do so through the most compressed (lowest-rank) representations, analogous to the retina’s compression of 10^8 photoreceptors into 10^6 optic nerve fibres.
- Connect the robustness of certain layers to compression with “cognitive noise tolerance”: which internal representations are robust to the kind of lossy compression that biological neural systems must endure?
- Investigate whether the outlier layers (e.g., layer 2 in Qwen3-0.6B with rank ≈ 1) serve a functional role analogous to specific cognitive bottlenecks.

13 Conclusion

The current experimental pipeline represents a well-engineered, honest evaluation of effective rank as a diagnostic for LLM compression. The key positive findings are:

1. Activation-aware SVD dramatically outperforms plain SVD, validating the importance of considering activation statistics in compression.
2. A strong, consistent negative correlation between effective rank and output energy exists across all four models tested, suggesting a fundamental structure in how transformer layers allocate representational capacity.
3. The effective rank profile reveals meaningful layer-level structure that connects naturally to information-theoretic concepts.

The key negative finding—that effective-rank-guided layerwise allocation does not outperform uniform allocation—is itself informative. It suggests that effective rank alone is insufficient as a compression guide, and that the interaction between spectral structure, activation magnitudes, and error propagation across layers requires a more sophisticated treatment. The most promising next steps are per-layer sensitivity analysis (to validate effective rank as a predictor rather than a budget tool) and deeper information-theoretic framing (to connect the empirical findings to the IB framework central to this course).

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